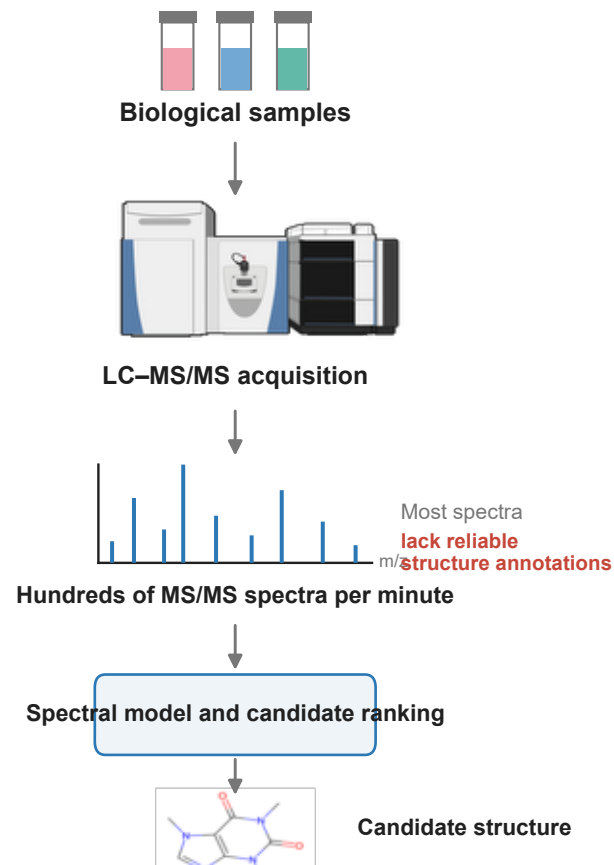
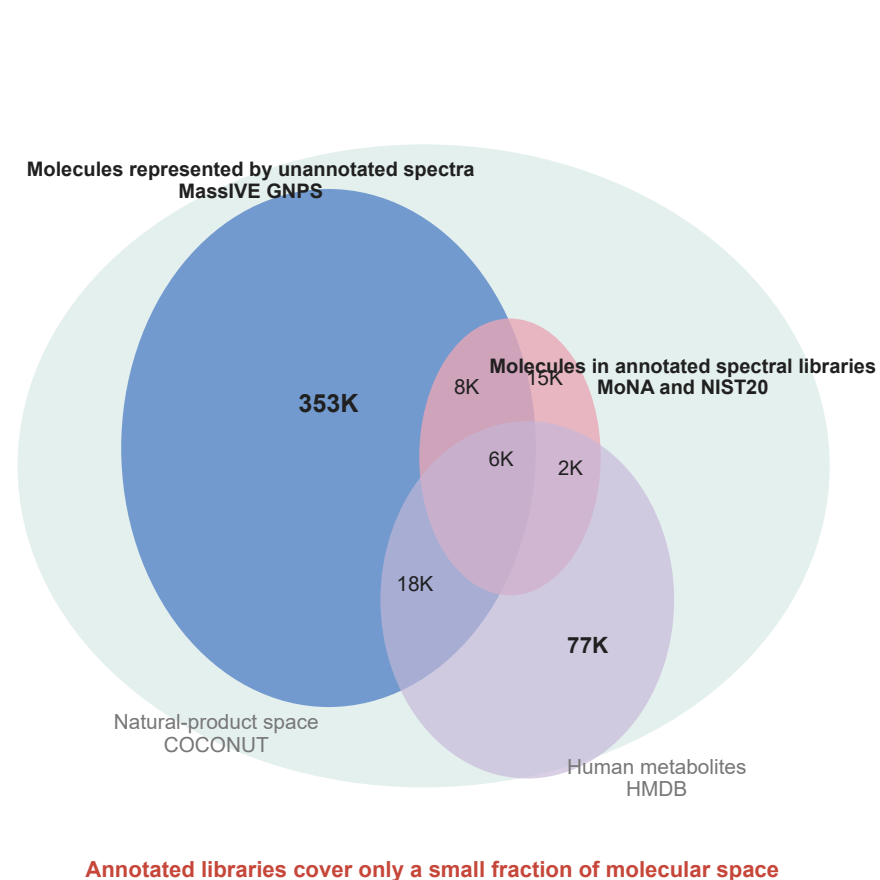


Figure 1. The core bottleneck in untargeted LC–MS/MS metabolite annotation

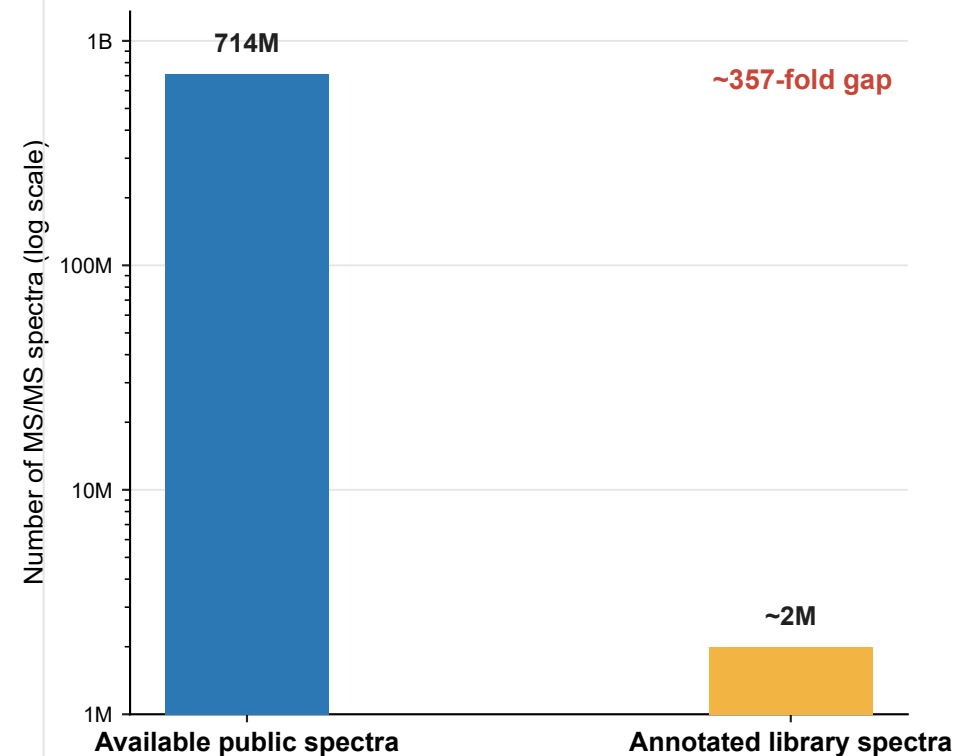
a From biological samples to candidate structures



b Annotated libraries cover limited molecular space



c Public spectra vastly outnumber annotated library spectra



Fewer than 10% of small-molecule MS/MS spectra can be reliably annotated